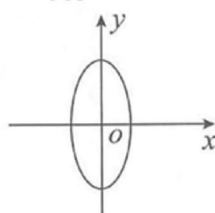


圆锥曲线综合习题册

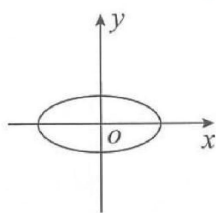
Conic Sections Comprehensive Exercise Book

一、选择题 (Multiple Choice Questions)

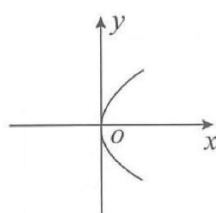
1. 圆心是 $(2, -3)$, 半径是 4 的圆的方程是 (). The equation of the circle with center $(2, -3)$ and radius 4 is ().
- A. $(x - 2)^2 + (y + 3)^2 = 4$
B. $\frac{x^2}{16} + \frac{y^2}{9} = 1$
C. $(x - 2)^2 + (y + 3)^2 = 16$
D. $(x + 2)^2 + (y - 3)^2 = 16$
2. 圆的一般方程是 $x^2 + y^2 - 6x + 4y - 12 = 0$, 它的圆心是 (). The general equation of the circle is $x^2 + y^2 - 6x + 4y - 12 = 0$, its center is ().
- A. $(3, -2)$
B. $(-3, 2)$
C. $(2, -3)$
D. $(-2, 3)$
3. 下列是椭圆方程的是 (). Which of the following is the equation of an ellipse? ().
- A. $\frac{x^2}{5} + \frac{y^2}{10} = 1$
B. $\frac{x^2}{25} - \frac{y^2}{16} = 1$
C. $x^2 + y^2 = 16$
D. $y^2 = -20x$
4. 椭圆 $\frac{x^2}{100} + \frac{y^2}{36} = 1$ 的焦点坐标是 (). The focal coordinates of the ellipse $\frac{x^2}{100} + \frac{y^2}{36} = 1$ are (). A. $(\pm 8, 0)$ B. $(0, \pm 8)$ C. $(\pm 14, 0)$ D. $(0, \pm 14)$
5. 椭圆 $\frac{x^2}{36} + \frac{y^2}{169} = 1$ 的图像是 (). The graph of the ellipse $\frac{x^2}{36} + \frac{y^2}{169} = 1$ is ().



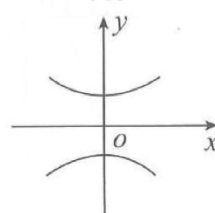
A.



B.



C.



D.

6. 下列是焦点在 y 轴的双曲线的方程是 (). Which of the following is the equation of a hyperbola with foci on the y-axis? ().
- A. $\frac{y^2}{36} - \frac{x^2}{16} = 1$
B. $\frac{y^2}{36} + \frac{x^2}{16} = 1$
C. $\frac{x^2}{36} - \frac{y^2}{16} = 1$
D. $\frac{x^2}{36} + \frac{y^2}{16} = 1$

7. 下列是双曲线方程的是 (). Which of the following is the equation of a hyperbola? ().

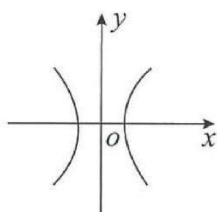
A. $x^2 + y^2 = 25$

B. $x = 3y^2$

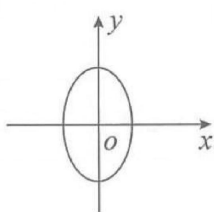
C. $\frac{x^2}{16} + \frac{y^2}{9} = 1$

D. $\frac{x^2}{16} - \frac{y^2}{9} = 1$

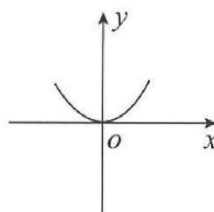
8. $\frac{y^2}{49} - \frac{x^2}{144} = 1$ 的图像是 (). The graph of $\frac{y^2}{49} - \frac{x^2}{144} = 1$ is ().



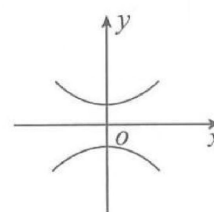
A.



B.



C.



D.

9. 顶点在原点, 焦点在 x 轴上, 并且经过点 $P(3, 6)$ 的抛物线方程是 (). The equation of the parabola with vertex at the origin, foci on the x-axis, and passing through point $P(3, 6)$ is ().

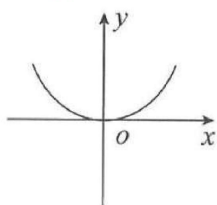
A. $y^2 = 12x$

B. $y^2 = 6x$

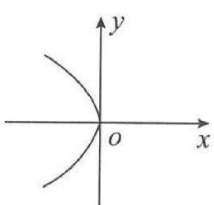
C. $x^2 = 3y$

D. $x^2 = \frac{1}{3}y$

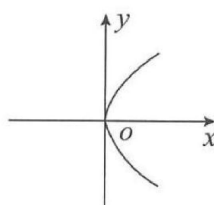
10. $x^2 = -8y$ 的图像是 (). The graph of $x^2 = -8y$ is ().



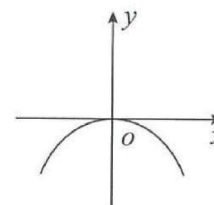
A.



B.

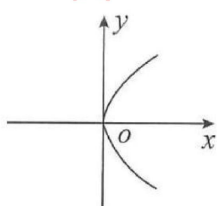


C.

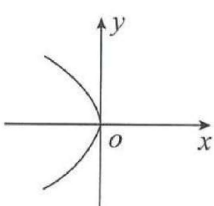


D.

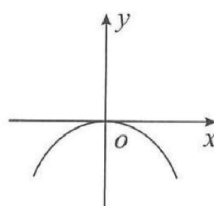
11. $y^2 = -20x$ 的图像是 (). The graph of $y^2 = -20x$ is ().



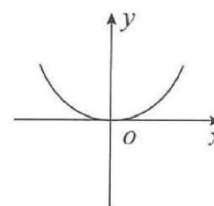
A.



B.



C.



D.

12. 抛物线 $y^2 = 6x$ 的开口方向是 (). The opening direction of the parabola $y^2 = 6x$ is ().

A. 向上 (Upward)

B. 向下 (Downward)

C. 向左 (To the left)

D. 向右 (To the right)

13. 抛物线 $x^2 = 12y$ 的开口方向是 (). The opening direction of the parabola $x^2 = 12y$ is ().

- A. 向上 (Upward)
- B. 向下 (Downward)
- C. 向左 (To the left)
- D. 向右 (To the right)

14. 焦点在 x 轴上的椭圆方程是 (). The equation of the ellipse with foci on the x-axis is ().

- A. $\frac{x^2}{16} - \frac{y^2}{9} = 1$
- B. $\frac{x^2}{16} + \frac{y^2}{9} = 1$
- C. $\frac{x^2}{9} + \frac{y^2}{16} = 1$
- D. $\frac{y^2}{16} - \frac{x^2}{9} = 1$

15. 焦点在 y 轴上的椭圆方程是 (). The equation of the ellipse with foci on the y-axis is ().

- A. $\frac{x^2}{9} + \frac{y^2}{36} = 1$
- B. $\frac{x^2}{9} - \frac{y^2}{36} = 1$
- C. $\frac{x^2}{36} + \frac{y^2}{16} = 1$
- D. $\frac{y^2}{36} - \frac{x^2}{9} = 1$

16. 焦点在 x 轴上的双曲线方程是 (). The equation of the hyperbola with foci on the x-axis is ().

- A. $\frac{x^2}{9} + \frac{y^2}{36} = 1$
- B. $\frac{x^2}{9} - \frac{y^2}{36} = 1$
- C. $\frac{x^2}{36} + \frac{y^2}{16} = 1$
- D. $\frac{y^2}{9} - \frac{x^2}{36} = 1$

17. 焦点在 y 轴上的双曲线方程是 (). The equation of the hyperbola with foci on the y-axis is ().

- A. $\frac{x^2}{16} + \frac{y^2}{9} = 1$
- B. $\frac{y^2}{16} + \frac{x^2}{9} = 1$
- C. $\frac{x^2}{9} - \frac{y^2}{16} = 1$
- D. $\frac{y^2}{9} - \frac{x^2}{16} = 1$

18. 抛物线 $x^2 = -12y$ 的焦点坐标是 (). The focal coordinates of the parabola $x^2 = -12y$ are ().

- A. (3, 0)
- B. (-3, 0)
- C. (0, 3)
- D. (0, -3)

19. 抛物线 $y^2 = 8x$ 的焦点坐标是 (). The focal coordinates of the parabola $y^2 = 8x$

are ().

A. (2, 0)

B. (-2, 0)

C. (0, 2)

D. (0, -2)

二、填空题 (Fill-in-the-Blank Questions)

1. $(x-3)^2 + (y-4)^2 = 16$ 的圆心是____; 半径是____. The center of $(x-3)^2 + (y-4)^2 = 16$ is ____; the radius is ____.

2. $(x+5)^2 + (y-2)^2 = 25$ 的圆心是____; 半径是____. The center of $(x+5)^2 + (y-2)^2 = 25$ is ____; the radius is ____.

3. $x^2 + (y+4)^2 = 36$ 的圆心是____; 半径是____. The center of $x^2 + (y+4)^2 = 36$ is ____; the radius is ____.

4. $x^2 + y^2 + 6x + 10y - 2 = 0$ 的圆心是____; 半径是____. The center of $x^2 + y^2 + 6x + 10y - 2 = 0$ is ____; the radius is ____.

5. $x^2 + y^2 - 8x - 4y - 5 = 0$ 的圆心是____; 半径是____. The center of $x^2 + y^2 - 8x - 4y - 5 = 0$ is ____; the radius is ____.

6. 椭圆 $\frac{x^2}{36} + \frac{y^2}{16} = 1$ 的焦点是____; 顶点是____; 离心率是____. The foci of the ellipse $\frac{x^2}{36} + \frac{y^2}{16} = 1$ are ____; the vertices are ____; the eccentricity is ____.

7. 椭圆 $\frac{x^2}{225} + \frac{y^2}{49} = 1$ 的长轴长是____; 短轴长是____. The length of the major axis of the ellipse $\frac{x^2}{225} + \frac{y^2}{49} = 1$ is ____; the length of the minor axis is ____.

8. 椭圆 $\frac{x^2}{16} + \frac{y^2}{25} = 1$ 的焦点是____; 焦距是____; 离心率是____. The foci of the ellipse $\frac{x^2}{16} + \frac{y^2}{25} = 1$ are ____; the focal length is ____; the eccentricity is ____.

9. 椭圆 $\frac{x^2}{25} + \frac{y^2}{49} = 1$ 的焦点是____; 焦距是____; 长轴长是____. The foci of the ellipse $\frac{x^2}{25} + \frac{y^2}{49} = 1$ are ____; the focal length is ____; the length of the major axis is ____.

10. 椭圆 $\frac{y^2}{49} + \frac{x^2}{9} = 1$ 的焦点是____; 顶点是____; 离心率是____. The foci of the ellipse $\frac{y^2}{49} + \frac{x^2}{9} = 1$ are ____; the vertices are ____; the eccentricity is ____.

11. 双曲线 $\frac{x^2}{25} - \frac{y^2}{16} = 1$ 的焦点是____; 离心率是____. The foci of the hyperbola $\frac{x^2}{25} - \frac{y^2}{16} = 1$ are ____; the eccentricity is ____.

12. 双曲线 $\frac{y^2}{16} - \frac{x^2}{36} = 1$ 的焦点是____. The foci of the hyperbola $\frac{y^2}{16} - \frac{x^2}{36} = 1$ are ____.

13. 双曲线 $\frac{x^2}{64} - \frac{y^2}{9} = 1$ 的焦点是____; 渐近线是____. The foci of the hyperbola $\frac{x^2}{64} - \frac{y^2}{9} = 1$ are ____; the asymptotes are ____.

14. 双曲线 $\frac{x^2}{16} - \frac{y^2}{25} = 1$ 的焦点是____; 顶点是____. The foci of the hyperbola $\frac{x^2}{16} - \frac{y^2}{25} = 1$ are ____; the vertices are ____.

15. 抛物线 $y^2 = -6x$ 的焦点坐标是____; 准线方程是____. The focal coordinates of the parabola $y^2 = -6x$ are ____; the directrix equation is ____.
16. 抛物线 $x^2 = 6y$ 的焦点坐标是____; 准线方程是____. The focal coordinates of the parabola $x^2 = 6y$ are ____; the directrix equation is ____.
17. 圆 $(x-4)^2 + (y+2)^2 = 9$ 的圆心是____; 半径是____. The center of $(x-4)^2 + (y+2)^2 = 9$ is ____; the radius is ____.
18. 圆 $x^2 + y^2 - 5x - 8y - 24 = 0$ 的圆心是____; 半径是____. The center of $x^2 + y^2 - 5x - 8y - 24 = 0$ is ____; the radius is ____.
19. 椭圆 $\frac{x^2}{25} + \frac{y^2}{49} = 1$ 的焦点是____. The foci of the ellipse $\frac{x^2}{25} + \frac{y^2}{49} = 1$ are ____.
20. 椭圆 $\frac{x^2}{18} + \frac{y^2}{8} = 1$ 的焦距是____. The focal length of the ellipse $\frac{x^2}{18} + \frac{y^2}{8} = 1$ is ____.
21. 双曲线 $\frac{x^2}{25} - \frac{y^2}{36} = 1$ 的焦点在____轴上. The foci of the hyperbola $\frac{x^2}{25} - \frac{y^2}{36} = 1$ are on the ____ axis.
22. 双曲线 $\frac{x^2}{36} - \frac{y^2}{16} = 1$ 的焦点在____轴上. The foci of the hyperbola $\frac{x^2}{36} - \frac{y^2}{16} = 1$ are on the ____ axis.
23. 抛物线 $y^2 = 10x$ 的开口方向是____. The opening direction of the parabola $y^2 = 10x$ is ____.
24. 抛物线 $x^2 = -10y$ 的开口方向是____; 焦点是____. The opening direction of the parabola $x^2 = -10y$ is ____; the focus is ____.

三、解答题 (Solution Questions)

1. 求以 $C(2, 5)$ 为圆心, 半径是 $\sqrt{10}$ 的圆的方程. Find the equation of the circle with center $C(2, 5)$ and radius $\sqrt{10}$.
2. 圆的方程是 $(x-5)^2 + (y+3)^2 = 16$, 求圆心和半径. The equation of the circle is $(x-5)^2 + (y+3)^2 = 16$, find its center and radius.
3. 圆的一般方程是 $x^2 + y^2 + 8x - 6y - 11 = 0$, 求圆心和半径. The general equation of the circle is $x^2 + y^2 + 8x - 6y - 11 = 0$, find its center and radius.
4. 写出下列圆的标准方程: Write the standard equations of the following circles: (1) 圆心过 $(3, -2)$, 半径是 6; (1) Center at $(3, -2)$ with radius 6; (2) 圆心过 $(0, 5)$, 半径是 3. (2) Center at $(0, 5)$ with radius 3.

5. 求顶点在原点, 焦点在 y 轴上, 并且经过点 $P(2, -4)$ 的抛物线方程. Find the equation of the parabola with vertex at the origin, foci on the y -axis, and passing through point $P(2, -4)$.
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